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Under review as a conference paper at ICAIS 2025

# EVALUATING THE TRADE-OFF BETWEEN PREDICTIVE ACCURACY AND SCREENING CAPACITY IN SOCIAL WELFARE PROGRAMS

**Anonymous authors**  
Paper under double-blind review

## ABSTRACT

As machine learning becomes integral to government programs aimed at identifying and assisting the most vulnerable populations, this paper investigates whether improving predictive accuracy is more beneficial than expanding screening capacity. We hypothesize that in typical operational conditions, enhancing capacity to reach more individuals will provide greater benefits than marginal gains in prediction accuracy. We introduce the Prediction-Access Ratio (PAR) to quantify this trade-off, guiding policymakers on when to invest in better models versus expanding access. Utilizing both mathematical modeling and a case study on long-term unemployment among German jobseekers, we demonstrate that expanding screening capacity generally leads to improved identification of the worst-off. Our findings empower policymakers with actionable insights, enabling more effective allocation of resources in equity-driven contexts.

## 1 INTRODUCTION

The effectiveness of social welfare programs is often contingent upon accurately identifying vulnerable populations. Traditional approaches have prioritized enhancing predictive models to accuracy. However, in resource-constrained environments, this focus may overlook criticalities to expand screening capacity. This paper explores a pivotal question: Is investing in capacity more impactful than marginal improvements in predictive accuracy? We introduce Prediction-Access Ratio (PAR), a novel metric that quantifies the trade-off between predictive accuracy and screening capacity expansion. We present findings from both mathematical and empirical case studies that highlight the importance of this trade-off, offering a framework for policymakers to optimize resource allocation in social welfare initiatives.

## 2 RELATED WORK

Existing literature emphasizes the significance of predictive models in welfare programs, particularly through causal simulation models (Irfiansyah et al., 2023; Orhan, 2024). However, there is a notable absence of comparative analyses focusing on the effectiveness of prediction accuracy versus screening capacity expansion. Our introduction of the PAR metric offers a fresh perspective on resource allocation in social programs, contrasting with prior work that primarily targets performance enhancement (Merluzzi et al., 2021; Ordóñez et al., 2025).

## 3 METHOD

The PAR is defined as the product of predictive accuracy and the number of individuals screened. It serves as a guide for policymakers to assess when to allocate resources toward enhancing predictive models versus increasing screening efforts. We employ both mathematical models and a case study to illustrate our findings. The mathematical component simulates various scenarios of predictive accuracy and screening capacity, while the empirical analysis leverages historical data from German jobseekers to validate the theoretical insights.

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## 4 EXPERIMENTAL SETUP

We conducted a series of experiments to evaluate the trade-offs inherent in resource allocation decisions. First, we simulated varying levels of predictive accuracy and screening capacity using mathematical models. Second, a case study analyzed historical data from jobseekers in Germany, assessing the impact of expanding screening capacity against improvements in predictive accuracy.

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 Summary

This paper investigates the critical trade-off between predictive accuracy and screening capacity in the context of social welfare programs, introducing the Prediction-Access Ratio (PAR) as a novel metric to quantify this balance. The core argument is that expanding screening capacity can often lead to greater improvements in identifying vulnerable populations than marginal gains in predictive accuracy, particularly in resource-constrained settings. The paper proposes that policymakers should consider this trade-off when allocating resources to optimize the impact of social welfare initiatives. The methodological approach involves both mathematical modeling and an empirical case study. The mathematical modeling component uses synthetic datasets with varying distributions and outlier percentages to simulate different scenarios and evaluate the impact of predictive accuracy and screening capacity on the identification of vulnerable individuals. The empirical component utilizes historical data from German jobseekers to validate the theoretical insights. The primary metric used to evaluate the performance of different scenarios is the PAR, which is calculated as the product of predictive accuracy and the number of individuals screened. The main finding from the mathematical modeling experiments is that expanding screening capacity can lead to significant improvements in the identification of vulnerable populations, even when predictive accuracy gains are marginal. This suggests that in resource-constrained settings, focusing on increasing the number of individuals screened may be a more effective strategy than solely focusing on improving the accuracy of the predictive model. The paper's overall significance lies in its introduction of the PAR metric and its emphasis on the importance of considering the trade-off between predictive accuracy and screening capacity in the design and implementation of social welfare programs. By highlighting this trade-off, the paper provides a valuable framework for policymakers to make more informed decisions about resource allocation and the impact of these programs on vulnerable populations. However, the paper's analysis is primarily based on mathematical modeling, and the empirical validation using real-world data is not as detailed as it could be. This limitation somewhat undermines the strength of the conclusions drawn from the study. Despite this, the core argument and the introduction of the PAR metric are valuable contributions to the field.

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 Strengths

The paper's primary strength lies in its identification of a crucial, yet often overlooked, aspect of social welfare program design: the trade-off between predictive accuracy and screening capacity. The introduction of the Prediction-Access Ratio (PAR) metric is a novel contribution that provides a quantifiable way to assess this trade-off. This metric, defined as the product of predictive accuracy and the number of individuals screened, offers a practical tool for policymakers to evaluate the effectiveness of different resource allocation strategies. The paper's focus on the potential benefits of expanding screening capacity, even with modest gains in predictive accuracy, is a valuable insight that challenges the conventional wisdom of solely focusing on model performance. The mathematical modeling approach, while not without its limitations, provides a systematic way to explore different scenarios and to simulate the impact of varying levels of predictive accuracy and screening capacity. The use of synthetic datasets with different distributions and outlier percentages allows for a robust analysis of the proposed metric under various conditions. The paper's emphasis on the practical implications of its findings for resource allocation in social welfare programs is another notable strength. By highlighting the potential for expanding screening capacity to improve the identification of vulnerable populations, the paper offers a valuable perspective that can inform policy decisions. The paper also attempts to validate its theoretical insights with an empirical case study using historical data from German jobseekers, although this aspect of the paper is not as fully developed as the mathematical modeling component. Despite the limitations in the empirical validation, the core argument of the paper and the introduction of the PAR metric are significant contributions that have the potential to influence how social welfare programs are designed and implemented. The paper's focus on a practical, real-world problem and its attempt to provide a quantifiable metric for evaluating solutions are commendable.

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 Weaknesses

After a thorough examination of the paper, I've identified several significant weaknesses that warrant careful consideration. First, while the paper introduces the Prediction-Access Ratio (PAR) metric, it lacks a detailed mathematical formulation and a comprehensive discussion of its theoretical properties. The paper defines PAR as the product of predictive accuracy and the number of individuals screened, and provides a basic mathematical expression:  $\text{PAR} = \text{Predictive Accuracy} \times \text{Number of Individuals Screened}$ . However, it fails to delve into the theoretical underpinnings of this metric. For instance, the paper does not discuss the range of possible PAR values, how the metric behaves under different conditions, or its sensitivity to variations in predictive accuracy and screening capacity. This lack of theoretical analysis makes it difficult to fully assess the validity and usefulness of the metric. The paper also does not explore the relationship between PAR and other established metrics in the field, which would have provided a better context for understanding its novelty and contribution. This omission is a significant weakness because it leaves the reader with an incomplete understanding of the metric's properties and limitations. My confidence in this assessment is high, as the paper's method section clearly defines the metric but does not provide the necessary theoretical analysis. Second, the paper's explanation of the case study on German jobseekers is limited and lacks essential details. While the paper mentions using historical data from German jobseekers to validate its theoretical insights, it fails to provide specific information about the dataset used, the features included, and the methodology for validating the findings. The paper states that predictive models, such as logistic regression and random forests, were trained on this data to predict long-term unemployment, and that the performance of these models was evaluated using metrics such as precision, recall, and F1-score. However, it does not provide any further details about the specific implementation of these models or the results obtained. Crucially, the paper does not present any empirical evidence from the case study to support its theoretical insights. The 'Experiments' and 'Results' sections focus primarily on the mathematical modeling experiments, and there is no clear presentation of the results of the case study. This lack of empirical validation significantly weakens the paper's conclusions, as it leaves the reader wondering whether the theoretical insights derived from the mathematical modeling hold true in real-world scenarios. The absence of details about the dataset, features, and validation methodology makes it difficult to assess the generalizability of the findings. My confidence in this assessment is high, as the paper explicitly mentions the case study but does not provide the necessary details or results. Third, the experimental design of the paper is inadequate to fully test the proposed PAR metric and the central hypothesis. The paper's 'Experiments' section describes two experiments: 'Mathematical Modeling of Predictive Accuracy vs. Screening Capacity' and 'Case Study on German Jobseekers'. However, the latter is not presented as a true experiment but rather as a validation of the mathematical modeling results. The 'Mathematical Modeling' experiment uses synthetic datasets with varying distributions and outlier percentages, and the metrics used are the number of vulnerable individuals identified and the PAR values. While the paper provides details about the purpose, baselines, metrics, and implementation of this experiment, it does not explicitly mention statistical significance testing or a discussion of the limitations of using simulated data. The lack of statistical significance testing makes it difficult to assess the robustness of the findings, and the absence of a discussion about the limitations of simulated data raises concerns about the generalizability of the results to real-world scenarios. Furthermore, the paper does not provide sufficient details about the baseline methods used for comparison, which makes it difficult to assess the relative performance of the proposed approach. The absence of a rigorous experimental design makes it difficult to validate the proposed metric and its practical implications. My confidence in this assessment is high, as the paper's 'Experiments' section lacks the necessary details about statistical significance, limitations of simulated data, and baseline comparisons. Finally, the paper's literature review is insufficient and does not adequately cover the relevant literature on resource allocation in social welfare programs and screening capacity. The 'RELATED WORK' section is brief and only cites three papers. While these papers are relevant to the topic, they do not provide a comprehensive overview of the existing literature or a detailed comparison of the proposed PAR metric to existing approaches. The paper's notable absence of comparative analyses focusing on the effectiveness of prediction accuracy versus screening capacity expansion, but this is likely due to a lack of thorough review of the relevant literature. The lack of a comprehensive literature review makes it difficult to contextualize the paper's contribution and to demonstrate the novelty of the proposed approach. My confidence in this assessment is high, as the paper's 'RELATED WORK' section is clearly limited in scope and depth.

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 Suggestions

To address the identified weaknesses, I recommend several concrete improvements. First, the authors should provide a more detailed mathematical formulation of the PAR metric, including a thorough discussion of its theoretical properties. This should include an analysis of the metric's range, its sensitivity to variations in predictive accuracy and screening capacity, and its relationship to other established metrics in the field. The authors should also explore the metric's behavior under different conditions and discuss its limitations. This deeper analysis will enhance the credibility of the metric and provide a more solid foundation for its use. Second, the authors should significantly expand the description of the case study on German jobseekers. This should include specific details about the dataset used, the features included, and the methodology for validating the findings. The authors should also present the empirical results of the case study, including the performance of the predictive models and the impact of varying screening capacities. This will provide much-needed empirical support for the paper's theoretical insights and enhance the generalizability of the findings. The authors should also discuss the limitations of the case study and how they might affect the conclusions. Third, the authors should improve the experimental design to better test the proposed PAR metric and the central hypothesis. This should include a more detailed description of the baseline methods used for comparison, the evaluation metrics, and the statistical significance of the results. The authors should also discuss the limitations of using simulated data and how they might affect the conclusions. For the mathematical modeling experiment, the authors should consider using a wider range of simulated datasets and conducting a more thorough analysis of the results. They should also explicitly mention the statistical tests used and report the p-values for all comparisons. Furthermore, they should discuss the effect size of their findings, which is a measure of the practical significance of the results. Fourth, the authors should expand the literature review to include more relevant work on resource allocation in social welfare programs, predictive modeling, and screening capacity. The authors should also discuss how the proposed PAR metric compares to existing approaches and how it addresses the limitations of previous work. This will provide a better context for understanding the paper's contribution and demonstrate the novelty of the proposed approach. The authors should also discuss the limitations of their literature review and how they might affect the novelty and significance of their proposed approach. Finally, the authors should consider restructuring the paper

to clearly separate the mathematical modeling component from the empirical case study. This will help to improve the clarity and organization of the paper and make it easier for the reader to follow the authors' arguments. By addressing these weaknesses, the authors can significantly improve the quality and impact of their paper.

? Questions

Based on my analysis, I have several questions that I believe are crucial for further clarification and understanding of the paper's contributions. First, could the authors provide a more detailed explanation of the theoretical properties of the PAR metric? Specifically, how does the metric behave under different conditions, what is its range, and how sensitive is it to variations in predictive accuracy and screening capacity? What is the relationship between PAR and other established metrics in the field? Second, could the authors provide more details about the dataset used in the case study on German jobseekers? What are the specific features included in the dataset, how was the data collected, and what is the size of the dataset? What specific machine learning models were used, and what were their hyperparameters? What were the results of the case study, and how do these results support the theoretical insights derived from the mathematical modeling? Third, could the authors elaborate on the experimental design of the mathematical modeling experiment? What specific statistical tests were used to assess the significance of the results, and what were the p-values for these tests? What was the rationale behind the choice of simulated datasets, and what are the limitations of using simulated data? What were the baseline methods used for comparison, and how did the proposed approach perform relative to these baselines? Fourth, could the authors provide a more comprehensive literature review that covers the relevant literature on resource allocation in social welfare programs, predictive modeling, and screening capacity? How does the proposed PAR metric compare to existing approaches in the literature, and how does it address the limitations of previous work? What are the limitations of the current literature review, and how might they affect the novelty and significance of the proposed approach? Finally, could the authors discuss the practical implications of their findings for policymakers? How can the PAR metric be used to inform resource allocation decisions in social welfare programs? What are the potential challenges and limitations of implementing the proposed approach in real-world settings? These questions are intended to address the key uncertainties and limitations identified in my analysis and to seek further clarification of the paper's core methodological choices and assumptions.

Scores

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Soundness: 1.5

Presentation: 1.25

Contribution: 1.25

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Rating: 2.5

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Summary

The paper studies the trade-off between improving predictive accuracy and expanding screening capacity in social welfare programs. It proposes the Prediction-Access Ratio (PAR), defined in Section 3 as the product of predictive accuracy and the number of individuals screened, as a guiding metric for resource allocation. The authors hypothesize that expanding capacity typically yields greater benefits than marginal accuracy gains, and claim support via mathematical modeling, a case study on long-term unemployment among German jobseekers, and a controlled experiment comparing model improvement versus capacity expansion (Abstract; Sections 3–4). The experiments section (Section 5) presents figures showing generic training loss/accuracy curves on synthetic datasets under different data distributions, outlier rates, and early stopping.

## ✓ Strengths

- Timely and practically relevant question: how to allocate scarce resources between improving models and expanding the reach of screening in social programs (Introduction).
- Clear, simple framing (PAR) that invites discussion of model performance versus program capacity (Section 3).
- Policy-oriented motivation and intent to support decision-making in equity-driven contexts (Abstract; Conclusion).

## ✗ Weaknesses

- Underspecified methodology and missing evidence: while the paper claims mathematical modeling, a German jobseeker case study, and a controlled experiment (Abstract; Sections 3–4), it provides no details on datasets, experimental protocols, baselines, or quantitative results.
- Figures 1–4 are generic ML training diagnostics on synthetic data (loss/accuracy over epochs under distribution shifts, outliers, early stopping) and do not compute or analyze the proposed PAR, nor do they evaluate screening capacity trade-offs or present the German case study.
- PAR is defined only as a product of accuracy and number screened (Section 3) without justification or linkage to decision-theoretic utility, costs, prevalence/base rates, ROC/PR characteristics, or thresholding; the metric risks being misleading.
- No operational definitions: predictive accuracy is not specified (e.g., AUROC, AUPRC, recall@k), and 'screening capacity' is not operationalized in units of cost/time/throughput; no budget constraints or cost models are introduced.
- No comparison to baselines or prior policy evaluation frameworks (e.g., cost-effectiveness analysis, value of information, constrained optimization) and limited related work specific to targeting/resource-allocation in social programs.
- Reproducibility is not possible: no data descriptions, code, model architectures, or precise procedures are provided.

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## ? Questions

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- Please formalize PAR. Why is the product of 'predictive accuracy' and 'number screened' the right functional form? Under what assumptions does PAR correspond to maximizing expected social welfare?
- How do you define 'predictive accuracy'? Is it AUROC, AUPRC, balanced accuracy, recall@k, top-k precision, calibration error, or a utility-weighted metric? How do you handle threshold selection?
- How is 'screening capacity' operationalized (e.g., number of caseworkers, available screening slots per period, cost per additional screen)? What are the budget and capacity constraints underpinning your analysis?
- Can you present the mathematical model promised in the Abstract and Section 3, including assumptions, cost/benefit functions, and the precise optimization problem? Please derive the decision boundary where investing in capacity dominates investing in accuracy.
- For the German jobseeker case study: What is the dataset, the outcome label (e.g., long-term unemployment after X months), prevalence, and timeframe? What models were used, how were they trained/evaluated, and what baselines were compared? Please report quantitative results that compute PAR (or a more principled utility metric) across interventions.
- For the controlled experiment (Section 4): What were the intervention arms, sample sizes, randomization protocol, outcome measures, and statistical analysis? What are the effect sizes and confidence intervals for identification of vulnerable individuals?
- How sensitive are your conclusions to prevalence, calibration, misclassification costs (false positives vs false negatives), heterogeneity of treatment effects, and diminishing returns to capacity expansion?
- How do fairness and equity considerations enter the objective? For instance, would maximizing PAR exacerbate disparities if accuracy differs across subgroups or if capacity expansion disproportionately reaches certain populations?

- Can you compare PAR with alternative metrics grounded in cost-effectiveness and welfare (e.g., expected utility per unit budget, net benefit, recall@budget, or constrained optimization on the PR curve)?

## ⚠ Limitations

- The current PAR definition ignores key factors for policy targeting: varying base rates, asymmetric costs, heterogeneous treatment effects, thresholding, calibration, and the shape of ROC/PR curves.
- Capacity expansion likely exhibits diminishing returns and queueing dynamics; the simple product form cannot capture congestion effects, outreach costs, or attrition.
- The analysis does not address selection bias or distribution shift between screened and unscreened populations.
- Potential negative societal impacts: expanding capacity without careful targeting may increase false positives, burdening individuals with unnecessary screening or stigmatization; conversely, relying on a single scalar metric can mask subgroup harms and equity issues.
- No discussion of privacy, data governance, or consent in the context of large-scale screening expansions.

## 🖼 Image Evaluation

Cross-Modal Consistency: 18/50

Textual Logical Soundness: 11/30

Visual Aesthetics & Clarity: 12/20

Overall Score: 41/100

Detailed Evaluation (≤500 words):

Visual ground truth (Image-first)

- Figure 1: Two panes.
- (a) Loss vs epochs; curves for Gaussian/Imbalanced/Uniform; large oscillations in Imbalanced.
- (b) Accuracy vs epochs; Imbalanced reaches ~0.9 with sharp dips; others rise slowly (~0.55–0.6).

Synopsis: Training dynamics across synthetic datasets.

- Figure 2: Two panes.
- (a) Loss vs epochs; Gaussian/Uniform/Exponential; gradual declines with spikes.
- (b) Accuracy vs epochs; all increase from ~0.5 to ~0.6.

Synopsis: Effect of data distribution on training dynamics.

- Figure 3: Six small panes.
- Loss and Accuracy for 0%, 5%, 10% outliers; similar saw-tooth loss; modest accuracy rises.

Synopsis: Outlier rate vs stability.

- Figure 4: Two panes.
- (a) Train/validation loss vs epochs; divergence then plateau.
- (b) Train/validation accuracy vs epochs; validation degrades/oscillates.

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Synopsis: Early stopping behavior.

1. Cross-Modal Consistency

- Major 1: Core visuals show only training loss/accuracy; no variables for access/screening capacity or PAR. Evidence: “The PAR is defined as the product of predictive accuracy and the number of individuals screened.” (Sec. 3); Figs. 1–4 contain only loss/accuracy vs epochs.
- Major 2: Promised German jobseeker case study and controlled two-arm comparison have no corresponding figures/tables. Evidence: “a case study analyzed historical data from jobseekers in Germany... a controlled experiment compared two intervention groups” (Sec. 4); no such results appear in Figs. 1–4.
- Minor 1: Fig. 1 caption claims “Imbalanced dataset shows instability,” yet right pane also shows sustained ~0.9 accuracy while others ~0.55, creating mixed messaging.
- Minor 2: Multiple uncited/uncaptioned images appear adjacent to Figs. 2–4, blurring what text refers to.

2. Text Logic

- Major 1: Flagship claim “expanding screening capacity generally leads to improved identification” lacks quantitative evidence or linkage to PAR. Evidence: “expanding screening capacity generally leads to improved identification of the worst-off” (Sec. 5/Conclusion); results show only generic training curves.
- Major 2: Broken sentence interrupts argument. Evidence: “Notably, the” (Sec. 5, preceding Fig. 3).
- Minor 1: PAR definition is stated but unanalysed (no normalization, bounds, or decision rule).

3. Figure Quality

- Minor 1: Sub-panes lack (a)/(b) labels; captions reference multi-panel content generically, hindering precise citation.
- Minor 2: Axes lack dataset sizes/units and whether accuracy is train or val (except Fig. 4).
- Minor 3: Some fonts near tick labels are small but still readable.

Key strengths:

- Plots are generally legible and stylistically consistent.
- Early-stopping figure contrasts train/validation behavior clearly.

Key weaknesses:

- Severe figure-text mismatch; no evidence for PAR, access expansion, or case study.
- Incomplete prose and missing experimental details.
- Multi-panel figures lack labels and operational context, failing the “figure-alone” policy relevance test.

Recommendations:

- Add figures/tables quantifying PAR vs. capacity/accuracy, plus German case-study outcomes.
- Provide subfigure labels, clear legends (train vs validation), and link every textual claim to a specific visual.

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 Scores

Originality: 2

Quality: 1

Clarity: 2

Significance: 2

Soundness: 1

Presentation: 2

Contribution: 2

Rating: 3



Summary

This paper explores the critical trade-off between predictive accuracy and screening capacity in the context of social welfare programs. The authors introduce the Prediction-Access Ratio (PAR) as a metric to quantify this trade-off, aiming to provide policymakers with a tool to optimize resource allocation. The core idea is that simply improving the accuracy of predictive models may not always be the most effective strategy; instead, expanding the reach of screening efforts might yield greater benefits in identifying vulnerable populations. The paper presents a mathematical model to simulate different scenarios of predictive accuracy and screening capacity, and it includes a case study using historical data from German jobseekers to validate the theoretical insights. The authors argue that in resource-constrained settings, focusing on expanding screening capacity may be more impactful than marginal improvements in predictive accuracy. The paper's methodology involves defining PAR as the product of predictive accuracy and the number of individuals screened, and then using mathematical modeling to explore the relationship between these two factors. The empirical component involves analyzing historical data from German jobseekers, although the specific details of this analysis are not extensively described in the provided text. The main findings suggest that expanding screening capacity can lead to substantial improvements in identifying vulnerable populations, even when predictive accuracy improvements are marginal. The authors conclude that the PAR framework can help policymakers make informed decisions about resource allocation in social welfare initiatives. While the paper presents an interesting and relevant problem, it suffers from several limitations, particularly in the clarity of its methodology, the depth of its empirical analysis, and the lack of comparison with existing methods. The paper's contribution lies in highlighting the importance of considering both predictive accuracy and screening capacity, but the practical implications and generalizability of the findings are not fully explored. The paper's overall significance is limited by the lack of detail in the case study and the absence of a thorough comparison with alternative approaches.

Strengths

The paper's primary strength lies in its identification of a crucial, yet often overlooked, aspect of social welfare program design: the trade-off between predictive accuracy and screening capacity. The authors correctly point out that focusing solely on improving predictive models may not be the most effective strategy, especially in resource-constrained environments. The introduction of the Prediction-Access Ratio (PAR) is a novel attempt to quantify this trade-off, providing a potential metric for policymakers to consider when allocating resources. The paper's focus on the practical implications of this trade-off is a notable strength, as it aims to provide actionable insights for those involved in designing and implementing social welfare programs. The use of a case study involving German jobseekers, although not detailed in the provided text, suggests an effort to ground the theoretical framework in a real-world context. The paper's attempt to bridge the gap between theoretical modeling and empirical analysis is a positive aspect, even though the execution of the empirical component could be improved. The paper's overall aim to provide a framework for optimizing resource allocation in social welfare initiatives is a valuable contribution, as it highlights the need to consider both the accuracy of predictive models and the reach of screening efforts. The paper's focus on the practical implications of the trade-off between predictive accuracy and screening capacity is also a notable strength, as it aims to provide actionable insights for policymakers. The paper's attempt to bridge the gap between theoretical modeling and empirical analysis is a positive aspect, even though the execution of the empirical component could be improved. The paper's overall aim to provide a framework for optimizing resource allocation in social welfare initiatives is a valuable contribution, as it highlights the need to consider both the accuracy of predictive models and the reach of screening efforts.

Weaknesses

After a thorough review of the provided paper and the reviewer comments, several significant weaknesses emerge. First, the paper lacks clarity in its methodology. The definition of PAR, while provided as the product of predictive accuracy and the number of individuals screened, is not thoroughly explained, and the reviewer raises a valid point about its limitations, particularly regarding the distribution of predictions. The paper does not explicitly state whether the goal is to maximize the number of true positives or to maximize the ratio of true positives to total individuals screened, leading to ambiguity in the interpretation of the results. The paper also fails to define key terms such as 'screening capacity' and 'worst-off' with sufficient precision, making it difficult to fully understand the scope and implications of the

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proposed framework. The lack of a clear objective function further compounds this issue, as it is unclear what the model is trying to optimize. Second, the paper's empirical analysis is weak. The case study using German jobseeker data is mentioned, but the details of this analysis are not provided in the text. The figures presented focus on training loss and accuracy curves for different synthetic datasets (Gaussian, Imbalanced, Uniform, Exponential) and outlier percentages, but these experiments do not directly demonstrate the application of PAR to the jobseeker data or compare the outcomes of different resource allocation strategies. The figures are also poorly labeled, with the x-axis consistently labeled as 'Epochs' without explaining what an 'epoch' represents in the context of the experiments. The y-axis labels are also confusing, using 'AUC' and 'Loss' without defining these terms or explaining their relevance to the specific problem. The lack of a detailed description of the empirical analysis and the poor labeling of the figures significantly undermine the paper's credibility. Third, the paper lacks a comparison with existing methods. The paper introduces PAR as a novel metric, but it does not compare its performance or insights with other existing methods for evaluating or optimizing social welfare programs. This lack of comparison makes it difficult to assess the relative merits of the proposed approach and its potential advantages over existing techniques. The paper also fails to discuss the limitations of the PAR metric and the potential challenges in its practical implementation. Fourth, the paper's motivation is not clearly articulated. While the paper states the problem it aims to address, it does not provide a detailed explanation of why this problem is important or what gaps in existing methods it aims to fill. The paper also does not clearly define the specific setting in which the proposed approach is intended to be used, making it difficult to assess its relevance and applicability. The paper's lack of a clear motivation and problem definition weakens its overall impact. Fifth, the paper's writing is poor. The text is not well-organized, and the arguments are not clearly presented. The lack of detail in the methodology and empirical analysis, combined with the poor labeling of the figures, makes the paper difficult to understand. The paper also contains several grammatical errors and typos, further detracting from its overall quality. Finally, the paper's contribution is limited. While the introduction of PAR is a novel attempt to quantify the trade-off between predictive accuracy and screening capacity, the paper does not fully explore the practical implications of this metric or its potential advantages over existing approaches. The lack of a thorough empirical analysis and the absence of a comparison with existing methods limit the paper's overall contribution to the field. The reviewer's concern about the simplicity of the model and the need for more real-world data is also valid, as the paper primarily relies on synthetic data for its experiments. The paper also does not adequately address the potential for false positives and false negatives, and how these errors might differentially impact the target population. The paper's failure to address these ethical considerations further limits its overall impact.

 **Suggestions**

To address the identified weaknesses, I recommend several concrete improvements. First, the authors should provide a more detailed and clear explanation of the PAR metric. This should include a clear definition of all terms, including 'screening capacity' and 'worst-off,' as well as a discussion of the assumptions underlying the metric, particularly regarding the distribution of predictions. The authors should also explicitly state the objective function that the model is trying to optimize, whether the goal is to maximize the number of true positives or to maximize the ratio of true positives to total individuals screened. Second, the authors should significantly expand the empirical analysis. This should include a detailed description of the German jobseeker dataset, the specific screening process used, and the outcomes measured. The authors should also provide a clear explanation of how PAR is calculated and used in the context of this case study. The figures should be clearly labeled, with the x and y axes explained in the captions. The authors should also consider using more real-world data and conducting experiments that directly compare the outcomes of different resource allocation strategies. Third, the authors should compare the performance of the proposed approach with existing methods for evaluating or optimizing social welfare programs. This should include a discussion of the advantages and disadvantages of the proposed approach compared to alternative techniques. The authors should also discuss the limitations of the PAR metric and the potential challenges in its practical implementation. Fourth, the authors should provide a more detailed explanation of the motivation behind the paper and the specific setting in which the proposed approach is intended to be used. This should include a discussion of the problem the paper aims to solve, why this problem is important, and what gaps in existing methods the paper aims to fill. Fifth, the authors should significantly improve the writing quality of the paper. This should include a more organized structure, clearer arguments, and fewer grammatical errors and typos. The authors should also ensure that all figures are clearly labeled and that all terms are defined. Sixth, the authors should address the ethical implications of using predictive models in social welfare programs. This should include a discussion of the potential for bias, the impact of false positives and false negatives, and the need for fairness and equity in the allocation of resources. Finally, the authors should consider expanding the scope of the paper to include a broader range of social welfare programs and contexts. This would help to demonstrate the generalizability of the proposed approach and its potential impact on a wider range of vulnerable populations. The authors should also consider the cost of screening and the cost of providing assistance, as these factors can significantly impact the optimal allocation of resources. The authors should also explore the sensitivity of the results to different assumptions about the costs and benefits of screening and assistance. The authors should also consider the potential for unintended consequences of expanding screening capacity, such as the stigmatization of individuals who are identified as potentially needing assistance.

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 **Questions**

Several key questions arise from my review of this paper. First, what are the specific assumptions underlying the mathematical model used to simulate different scenarios of predictive accuracy and screening capacity? How do these assumptions affect the generalizability of the findings? Second, how is the 'worst-off'

population defined in the context of the German jobseeker case study? What specific criteria were used to identify individuals as being among the 'worst-off'? Third, what is the practical interpretation of the PAR metric? How can policymakers use this metric to make concrete decisions about resource allocation? What are the limitations of this metric, and under what conditions might it be misleading? Fourth, what is the cost of screening an individual, and what is the cost of providing assistance to an individual? How do these costs affect the optimal allocation of resources? Fifth, what is the potential for bias in the predictive models used in the paper? How might this bias differentially impact different subgroups within the population? Sixth, what are the potential unintended consequences of expanding screening capacity? Could this lead to the stigmatization of individuals who are identified as potentially needing assistance? Seventh, what is the specific research question that the paper aims to answer? What is the gap in existing methods that the paper aims to fill? Finally, what is the meaning of an 'epoch' in the context of the experiments presented in the figures? What data is being presented in these figures, and how does it relate to the overall findings of the paper?

 Scores

Soundness: 1.0

Presentation: 1.0

Contribution: 1.0

Rating: 1

Keywords

Extract Keywords

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Insights

Click the button to extract insights